Deep Learning for Image Classification: A Convolutional Approach

Abstract
Image classification is a fundamental task in computer vision. In this paper we propose a novel convolutional neural network architecture called DeepNet that achieves state-of-the-art accuracy on the CIFAR-10 benchmark. Our approach uses residual connections and attention mechanisms to improve gradient flow during training.

Introduction
Convolutional neural networks have revolutionized computer vision over the past decade. Prior work by Krizhevsky et al. demonstrated the power of deep architectures on ImageNet. However, training very deep networks remains challenging due to vanishing gradients. We introduce a new method to address this limitation.

Methodology
We propose a residual attention block that combines skip connections with a lightweight self-attention module. Our method reduces the number of parameters by 30 percent compared to ResNet-50 while maintaining comparable accuracy. We use ImageNet pretraining followed by fine-tuning on CIFAR-10.

Results
Our experiments show that DeepNet achieves 95.2 percent accuracy on CIFAR-10, outperforming the ResNet-50 baseline by 2.1 percentage points. We also demonstrate faster convergence, requiring 40 percent fewer training epochs. Ablation studies confirm that both the residual connections and attention module contribute to the improved performance.

Conclusion
This paper presented DeepNet, a novel architecture combining residual connections and attention for image classification. Our contribution demonstrates that lightweight attention mechanisms can significantly improve both accuracy and training efficiency. Future work will explore extending this approach to object detection tasks.